

The Impact of Neurocognition and Symptom Severity on Community Functioning in Psychosis: A Transdiagnostic Analysis

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Background

- Evidence indicates that cognitive performance correlates with community functioning in both schizophrenia (SZ) and bipolar disorder with psychosis (BD). However, whether the effects of symptom severity and cognitive impairment on community functioning differ between SZ and BD is still unclear.
- Thus, we aimed to examine the relationship between cognition (MCCB), symptom severity (PANSS, YMRS, MADRS), and community functioning (MCAS) in 74 people with either SZ or BD, and to evaluate the extent to which cognitive and clinical predictors of community functioning differed across diagnoses.

Methods

- Cognition** was assessed with the MATRICS Consensus Cognitive Battery (MCCB Composite)
- Symptom severity:** Positive And Negative Syndrome Scale (PANSS Positive, Negative, General, Total), Young Mania Rating Scale (YMRS Total), Montgomery–Åsberg Depression Rating Scale (MADRS Total)
- Community functioning:** Multnomah Community Ability Scale (MCAS Total); the MCAS assess four main areas of functional ability: health, adaptation, social skills, and behavior in the community
- Statistical analyses** included Pearson's correlations and stepwise linear regression models. Age and education variables were controlled for.

Demographic, Cognitive, and Clinical Information

Table 1

Demographic, Cognitive, and Clinical Measures by Diagnosis

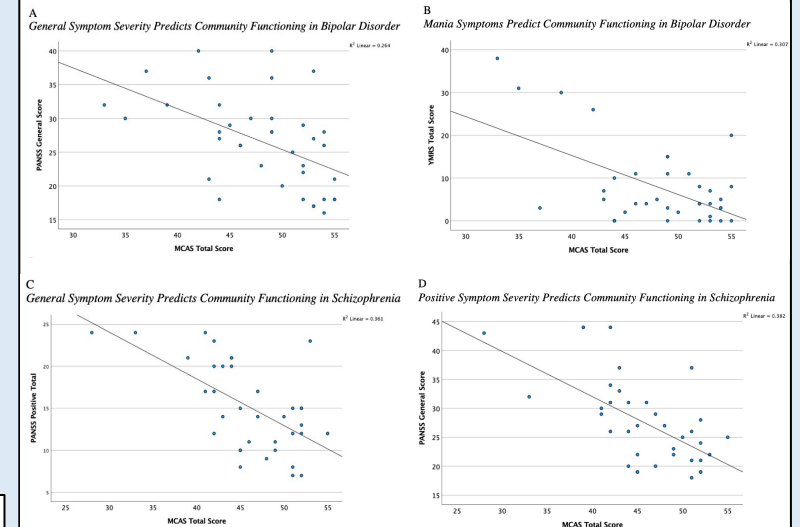
	SZ spectrum N = 36	BD N = 38	p-value
Sample size			
Sex (%)			.302
Female	30.56	42.11	
Male	69.44	57.89	
Mean age ± SD (years)	36.22 ± 12.46	26.34 ± 7.87	.000**
Race (%)			.083
Caucasian	77.78	92.11	
Other	22.22	7.89	
Mean education ± SD (years)	14.33 ± 1.76	15.47 ± 1.59	.002*
PANSS Scores			
Positive	15.17 ± 5.28	11.58 ± 5.81	.007*
Negative	13.69 ± 6.98	11.29 ± 3.93	.080
General	27.33 ± 7.22	26.55 ± 6.81	.634
Total	56.57 ± 16.27	49.42 ± 12.41	.019*
YMRS Total	8.58 ± 7.43	7.89 ± 9.52	.731
MCCB Composite	37.28 ± 9.68	46.24 ± 9.75	<.001**
MCAS Total	46 ± 5.71	48.08 ± 5.78	.124

* $p < .05$; ** $p < .001$

Results

- Cognition was correlated with community functioning in SZ ($R^2 = .123$, $F(1, 34) = 4.770$, $p = .036$) but not BD ($R^2 = .013$, $F(1, 36) = .475$, $p = .495$)
- Stepwise linear regression run separately by group including demographic, clinical, and cognitive measures as predictors of MCAS scores showed different predictive patterns by diagnosis:
 - BD:** several demographic and symptom variables were retained in the model ($R^2 = .649$, $F(6, 31) = 9.53$, $p < .0001$); however, only YMRS total and PANSS general were significant independent predictors (YMRS: $p = .001$; PANSS general: $p = .035$).
 - SZ:** only PANSS positive and PANSS general remained in the model ($R^2 = .519$, $F(2, 32) = 17.26$, $p < .0001$); both were significant independent predictors (PANSS positive: $p = .010$; PANSS general: $p = .004$).

Figure 1



Discussion

- We failed to find that cognition was an independent predictor of community functioning after accounting for symptom severity in our sample.
- Our results indicate that **different symptom profiles were predictive of community functioning** in SZ and BD. Positive symptoms were predictive of community functioning in SZ only, whereas PANSS general was predictive of community functioning in both SZ and BD and symptoms of mania were predictive of community functioning in BD. This suggests that, even in clinically stable outpatients, residual symptoms have significant and diagnosis-specific effects on community functioning.